SMART WITHDRAWAL AI

STRATEGIC MINDS

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Video

PRODUCT UNDERSTANDING & PROBLEM STATEMENT

The Problem: Financial Intelligence Comes Too Late.

Today
Portfolio → Dashboard → Insights → User Decides

Critical Moment: "Withdraw"
Investor needs to understand:
Portfolio Impact | Goal Impact | Risk Impact | Withdrawal Alternatives

The Gap
Information exists. Context is missing at the moment of decision.

Opportunity
Retrospective Information → Proactive Decision Support

INDUSTRY & MARKET RESEARCH

Investors Have Information — But Decision Context Is Missing

Investors seek information, but lack context at the moment of decision.

Key Findings | (SEBI Investor Survey 2025)



Report **risk & uncertainty** as a challenge



Uncertain whether to stay invested or exit



Seek **market** insights



Seek performance information

What Investors Seek



Competitor Analysis

Existing platforms offer valuable features, but the withdrawal decision remains fragmented and incomplete.

Full support | Partial support | Not available

Feature / Capability	Genvest	Vestmind	Novelty Wealth	MyFi	Groww	Stack Wealth
Portfolio Tracking & Analysis	✓	✓	✓	✓	✓	✓
Goals & Financial Planning	✓	✓	✓	✓	✓	◐
Risk Profiling & Allocation Advice	✓	✓	✓	✓	✓	◐
Scenario Analysis	◐	✓	✓	◐	◐	◐
Withdrawal / Liquidity Intelligence	◐	◐	◐	◐	◐	✓
Tax & Cost Insights	◐	◐	◐	◐	◐	✓
Decision-Moment Intervention	✗	✗	✗	◐	✗	◐
Personalized Recommendations	✓	✓	✓	✓	✓	◐

Key Takeaways

- Most platforms focus on portfolio, goals and risk.
- Withdrawal, tax and cost analysis are limited and fragmented.
- Decision-moment intervention is rare.
- There's an opportunity to bring all these capabilities together around the withdrawal decision.

TARGET USER PERSONA

Core Need
A personalized, explainable view of the consequences and trade-offs — before committing to the withdrawal.

Who Faces This Decision?
Goal-Oriented Retail Investor
Age: 25–35 | **Income:** ₹1L+/month | **Risk:** Moderate
Goal: Home / Major Goal | **Horizon:** 3–5 Years
Investment Context
Portfolio: ₹8–10L | **Withdrawal:** ~₹2L
Mix: Equity + Debt + Index/Other Assets
Decision Moment
Needs ₹2L → Clicks "Withdraw"
"What will this withdrawal change for me?"
Portfolio • Goal • Risk • Liquidity • Tax & Costs

PROBLEM DEFINITION & OPPORTUNITY

The Critical Gap: What Happens Before Withdrawal?

Current Journey

Need Money → Open Portfolio → Withdraw → Enter Amount → Confirm

Missing Layer

"What will this withdrawal change for me?"

Portfolio Impact
Goal Impact
Risk & Liquidity
Tax & Costs
Alternative Amounts

The Opportunity

Turn a Transaction into a Decision

PRODUCT VISION & VALUE PROPOSITION

How Smart Withdrawal AI Creates Value….??

UNDERSTAND

Portfolio • Goal • Risk • Tax

COMPARE

₹0 vs ₹1L vs ₹2L

EXPLORE

Redeem Now vs Stay Invested

DECIDE

Investor remains in control

Value Proposition

Before you withdraw, understand what you're changing — now and potentially in the future.

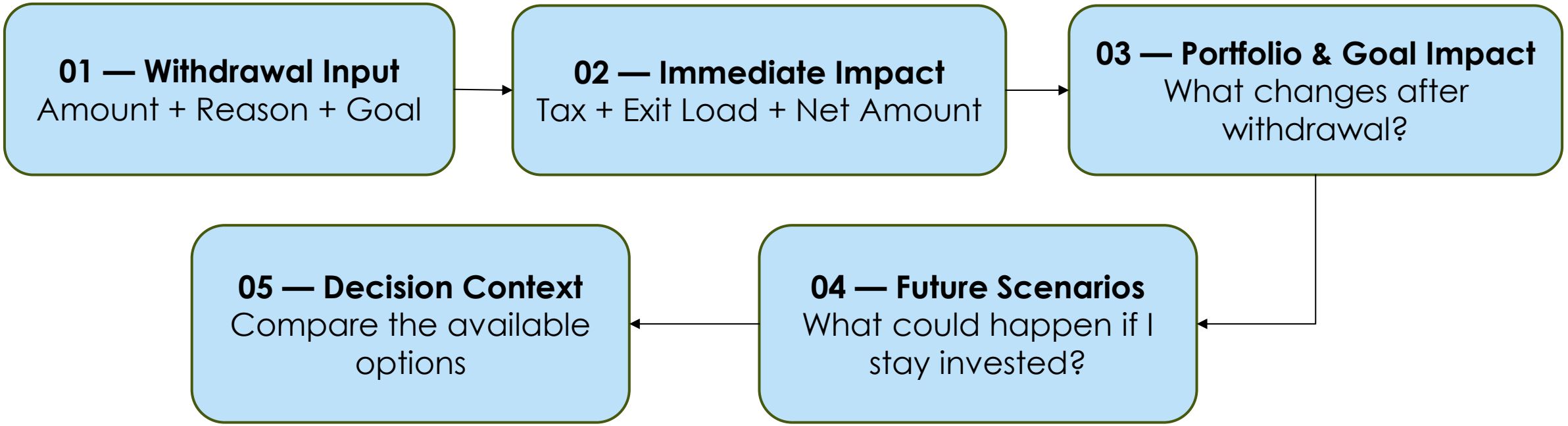
PRODUCT STRATEGY & INNOVATION

Portfolio Intelligence → Decision Intelligence Traditional
Portfolio → Information → User Interprets → Withdraw
Smart Withdrawal AI
Withdrawal Trigger
↓
AI Analysis
↓
Immediate Impact + Tax / Cost
↓
Redeem Now / Stay Invested
↓
Alternative Withdrawal Amounts
↓
Investor Decides

4 Innovation Pillars
01 — Decision-Moment
Intervenes before withdrawal.
02 — Personal Context
Portfolio + Goals + Risk + Liquidity.
03 — Scenario Analysis
Compare different withdrawal amounts.
04 — Explainable AI
Shows why an insight appears.

PRODUCT EXPERIENCE

One Withdrawal. Multiple Layers of Understanding.



USER DECIDES: AI informs. It does not make the decision.

SCENARIO ANALYSIS

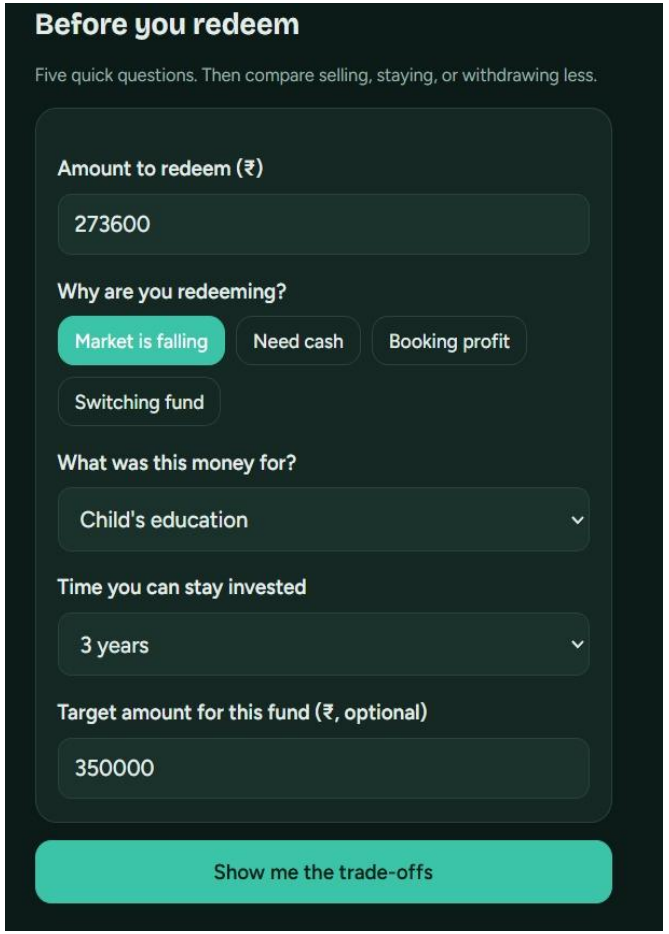
If You Redeem Now
₹2,00,000 Withdrawal
Gross Redemption: ₹2L
Tax + Exit Load: ₹X
Amount Received: ₹X
Portfolio: ₹8.42L → ₹6.42L
Goal Impact: Timeline may change
Tax & cost are estimates.

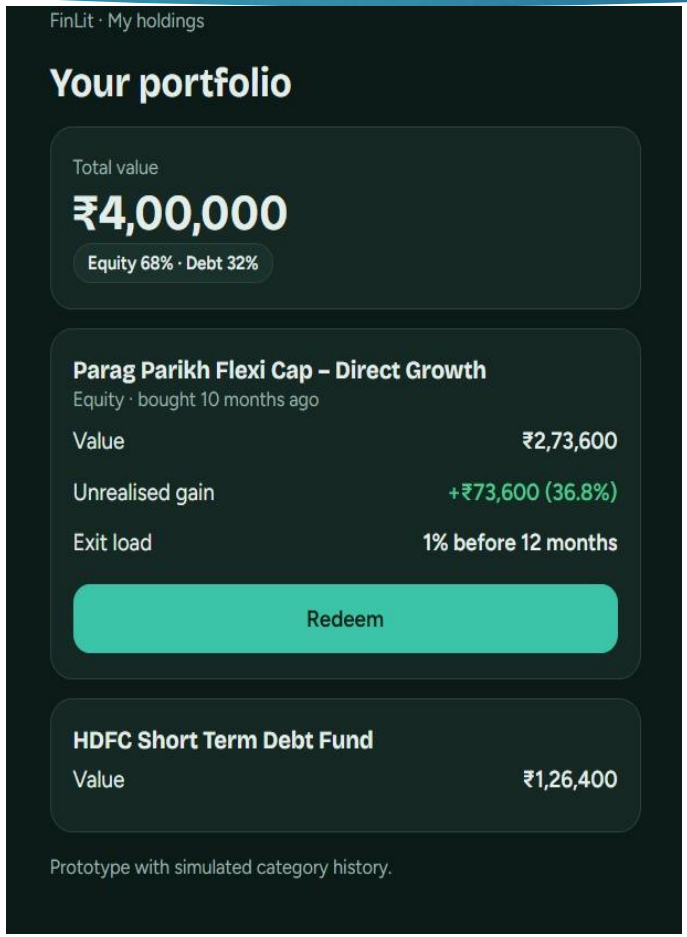
If You Stay Invested
What could happen?
Historical Returns
Volatility & Drawdown
Future Outcome Range: **P10 | P50 | P90**
Goal: **₹15L in 4 Years**
Goal Probability: XX%
Model-based scenarios, not guaranteed returns.
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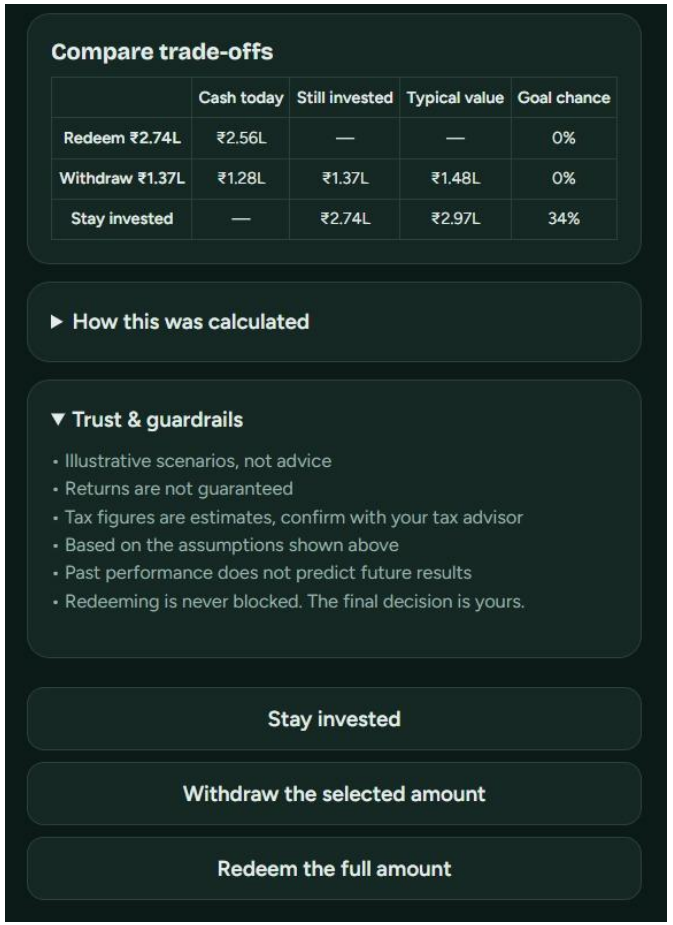
CLICKABLE PROTOTYPE

prototype

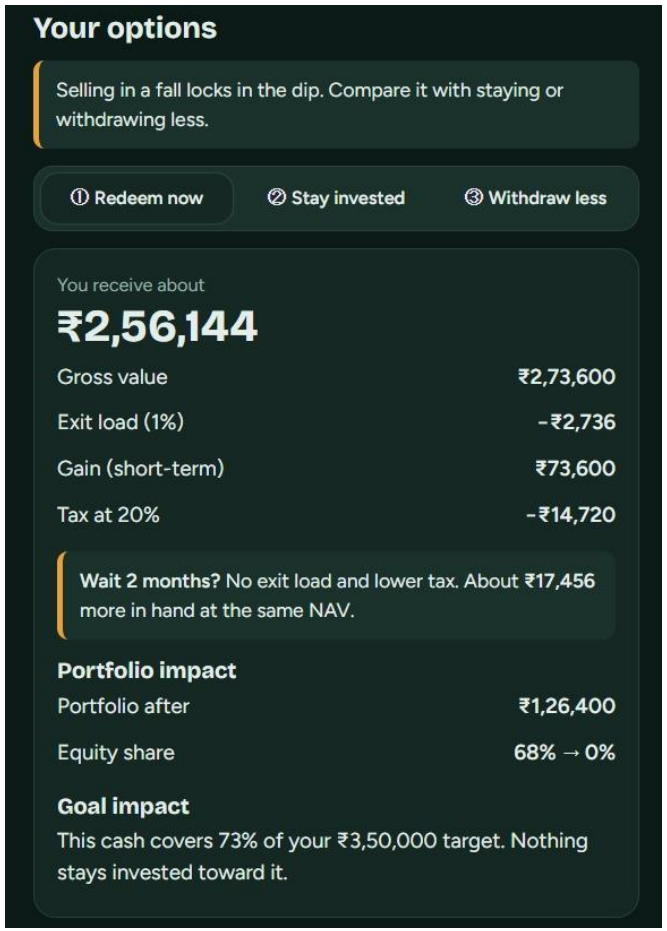
Video

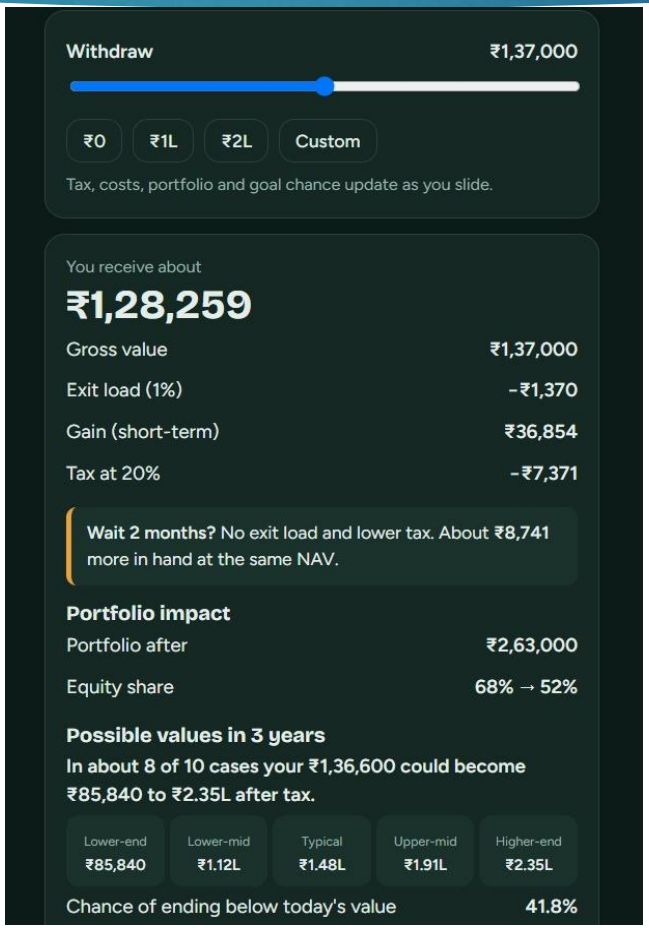


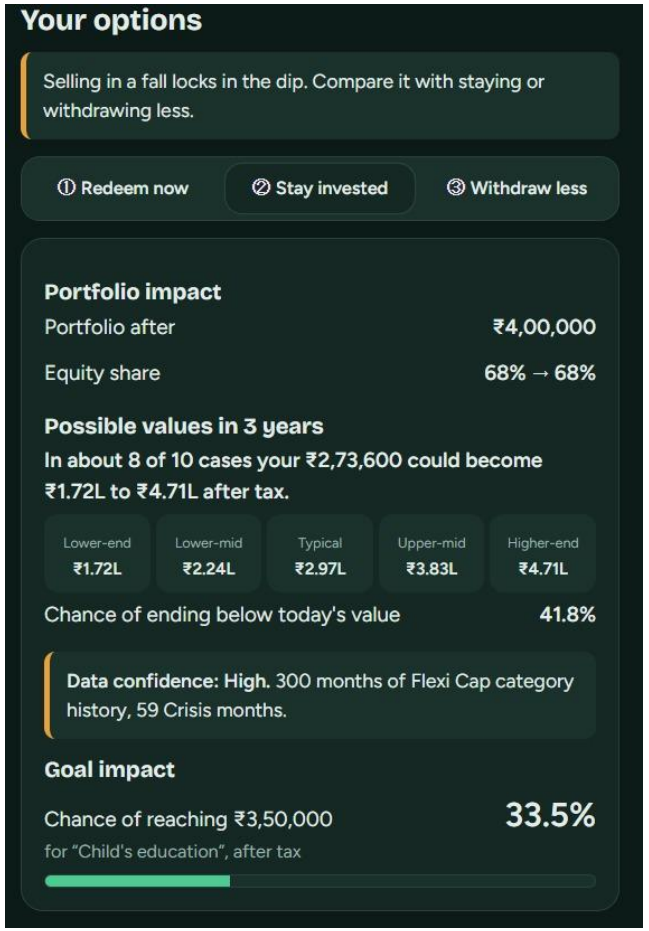




CLICKABLE PROTOTYPE







FINANCIAL ANALYSIS

Regime-Aware Monte Carlo Simulation

1. Historical Market Analysis: Analyze historical NAV data to identify returns, volatility, and drawdowns

2. Market Regime Identification: Classify market conditions into four regimes: Bull | Normal | Bear | Crisis

3. Transition Matrix: Estimate the probability of moving from one market regime to another using historical data.

4. Monte Carlo Simulation: 10,000 Paths: Current Regime → Next Regime → Historical Return → Portfolio Value *Repeated across the investment horizon.*

5. Future Outcome Analysis: Analyze the 10,000 simulated paths to estimate a range of possible future portfolio values.

6. Goal Achievement Probability: Measure the proportion of simulated paths that reach the investor's target goal.

PRODUCT EXECUTION

MVP
Must Have
Withdrawal trigger
Portfolio analysis
Tax / cost estimate
Goal impact
₹0 / ₹1L / ₹2L scenarios
Redeem vs Stay Invested
Explainable insights

Next
Multiple Goals
More Investment Products
Deeper Personalization
Advanced Scenarios

Success Metrics
North Star:
% of users who review decision context before withdrawal
Supporting:
Scenario Exploration • Analysis Completion • User Confidence

EXPLAINABLE AI

AI That Explains — Not Just Recommends
Why am I seeing this?
We looked at:
Portfolio
Goal
Withdrawal Amount
Investment History
Tax / Cost
↓
What does it mean?
"Redeeming ₹2L reduces the amount currently invested toward your goal.
What are the assumptions?
Historical Data • Time Horizon • Return Assumptions • Tax Inputs
AI informs. Investor decides.

Click here:

Prototype

Video

REFERENCES:
Hamilton (1989), Regime-Switching Models; Graf & Korn (2020), Monte Carlo Simulation for Risk-Return Analysis;
Detemple et al. (2003), A Monte Carlo Method for Optimal Portfolios.
SEBI_Investor_Survey_2025_Research Notes_Investment_Withdrawal.
Investment Withdrawal Core Research Data